



SHUWEN GONG

System Modelling & Control · Vehicle Dynamics & Powertrain

PROFILE

Mobility Engineering M.Sc. at Chalmers, focused on vehicle system modelling and control in MATLAB/Simulink and on identifying models from measured data. Available for a 30 ECTS thesis from January 2027. Earlier: fighter-aircraft final assembly (3 yrs) and live-service game systems design (4.5 yrs) — the habit of looking at the whole system before changing a part.

CONTACTS

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PORTFOLIO
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SKILLS

Control & Identification

State feedback, pole placement
Observer design, discretisation
System identification
Signal processing

Modelling & Simulation

MATLAB, Simulink, QSS
Vehicle dynamics, quarter-car
Hybrid powertrains, driving cycles

Data & Software

IMU data, PSD, filtering
Python, Git

Systems & Process

PFMEA, PDM/ERP/CAPP, CATIA

Languages

English (professional), Chinese (native), Swedish (learning)

EDUCATION

M.Sc. Mobility Engineering

Chalmers University of Technology

Sep 2025 – Jun 2027 (expected)

Gothenburg, Sweden

- Coursework: Systems and Mechatronics for Mobility Engineering; Electric and Hybrid Vehicles; Propulsion and Energy Systems for Transport; Vehicle and Traffic Safety; Connected Fleets in Data-Driven Engineering.
- Reversing aid for a tractor–semitrailer: pole-placement state feedback with a discretised observer, validated on the full nonlinear model across trailer lengths and speeds (Simulink).
- Series and parallel HEV powertrain design for a Class M reference vehicle: component sizing from top-speed and acceleration requirements, battery state-of-charge management, engine operating-point placement over NEDC and a custom cycle (QSS).
- Ride comfort on a Volvo 7900 Electric: bus IMU data → road-excitation PSD → quarter-car model; suspension stiffness and damping optimised to cut body-acceleration RMS under deflection and tyre-force constraints (MATLAB).
- Cyclist-overtaking field study with a LIDAR/IMU/GPS-instrumented bicycle: measured the lateral clearance drivers leave and how oncoming traffic compresses it.

B.Eng. Aircraft Manufacturing Engineering

Nanjing University of Aeronautics and Astronautics (NUAA)

Sep 2013 – Jun 2017

Nanjing, China

- Coursework in dynamics, control fundamentals, and engineering mathematics.

WORK EXPERIENCE

Production Engineer, Aircraft Final Assembly

AVIC Chengdu Aircraft Industrial (Group) Co., Ltd.

Jul 2017 – May 2020

Chengdu, China

- Process planning and PFMEA for fighter-aircraft final-assembly stages; coordinated mechanical, avionics and quality teams across multi-stage integration. Led on-site technical support in Pakistan for a new aircraft model deployment.
- First inventor, granted utility-model patent CN 210090631 U: a portable weight-on-wheels switch inspection device for a flight-critical final-assembly check.

Game Systems Designer

Aoshitang · Zlong Games · Duoyi Network

Aug 2020 – Feb 2025

China

- Owned economy and progression rule systems, feature roadmaps and live-ops planning for four shipped live-service games reaching hundreds of thousands of players.
- Defined KPIs and tuned rule systems against player behaviour data after each release.

ADDITIONAL EXPERIENCE

Psychological Support Volunteer | KnowYourself

China, Jul – Dec 2019